

MEDIATEK

MT8668 GNSS Sepcification

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Version History

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1 Introduction

1.1 Purpose

The specifications of MediaTek automotive GNSS module, MT8668, are provided in this document.

1.2 Scope

The supported functions, performance, and message output of MediaTek automotive GNSS module, MT8668, are detailed in this document.

[Chapter 1](#) Presents the purpose and scope of the document.

[Chapter 2](#) Lists the abbreviations.

[Chapter 3](#) Introduces the supported functions.

[Chapter 4](#) Introduces the performance.

[Chapter 5](#) Introduces the message output.

[Chapter 6](#) Lists the related documents.

2 Abbreviations

Table 2-1. Abbreviations

Abbreviation	Explanation
ADR	Automotive Dead Reckoning
A-GNSS	Assisted Global Navigation Satellite System
AIC	Active Interference Cancellation
CATARC	China Automotive Technology and Research Center
C/N0	Carrier-to-Noise Density Ratio
EDT	Error of Distance Travelled
EGNOS	European Geostationary Navigation Overlay Service
EPO	Extended Prediction Orbit
GAGAN	GPS Aided GEO Augmented Navigation
MSAS	Multi-functional Satellite Augmentation System
NLP	Network Location Provider
NTP	Network Time Protocol
PPS	Pulse-Per-Second
RTK	Real-time Kinematic Positioning
SDCM	System for Differential Corrections and Monitoring
SPP	Single Point Positioning
SUPL	Secure User Plane Location
T _a	Ambient Temperature
TCXO	Temperature Compensated Crystal Oscillator
TTFF	Time To First Fix
UTC	Coordinated Universal Time
WAAS	Wide Area Augmentation System
WGS84	World Geodetic System 1984

3 Function

3.1 RF Chip

MT8668 supports with the RF chip of MT6637.

3.2 Positioning Method

Table 3-1. Supported positioning method

Function	Positioning Method
Supported positioning method	SPP (Single Point Positioning)
	ADR (Automotive Dead Reckoning) ⁽¹⁾

(1) Depends on software version.

3.3 Constellation

Table 3-2. Supported configured GNSS constellation

Function	GNSS Constellation
Supported configured GNSS constellations	BeiDou-Only ⁽¹⁾
	GPS + GLONASS + Galileo + BeiDou + NavIC

(1) Conforms to latest CATARC specifications and test scenarios.

3.4 Satellite Signal

Table 3-3. Supported satellite signal baseband frequency

Signal		Baseband Frequency
GPS	L1C/A	1575.420 MHz
	L5	1176.450 MHz
Galileo	E1	1575.420 MHz
	E5a	1176.450 MHz
GLONASS	L1OF	1598.0625 to 1609.3125 MHz
BeiDou	B1I	1561.098 MHz
	B2a	1176.450 MHz
QZSS	L1C/A	1575.420 MHz
	L1S	1575.420 MHz
	L5	1176.450 MHz
NavIC/IRNSS	L5	1176.450 MHz

3.5 Satellite-Based Augmentation System (SBAS)

Table 3-4. Supported SBAS

Function	SBAS
Supported SBAS	WAAS (USA)
	SDCM (Russia)
	EGNOS (Europe)
	MSAS (Japan)
	GAGAN (India)

3.6 One Pulse-Per-Second (1PPS)

MT8668 supports the function of one pulse-per-second (1PPS) output rate.

3.7 Active Interference Cancellation (AIC)

MT8668 supports the function of AIC.

3.8 Coordinate System

MT8668 supports the coordinate system of WGS84.

3.9 Anti-jamming

MT8668 supports the function of jamming detection and filtering.

Note:

- Only supports CW tone detection and filtering.

3.10 A-GNSS

Table 3-5. Supported A-GNSS functions

Function	A-GNSS Function
Supported A-GNSS functions	A-GNSS (GPS, BDS, GLO, GAL) ⁽¹⁾
	EPO
	E911
	SUPL 2.0

(1) Supported constellations depend on service provider.

3.11 eCall

MT8668 supports EU eCall GNSS part and ERA-GLONASS(GOST) GNSS part.

3.12 Oscillator

MT8668 supports with the oscillator of TCXO.

4 Performance

4.1 Noise Figure

MT8668 supports performance in this datasheet with noise figure ≤ 2.5 dB.

4.2 Max Tracking Channel

MT8668 supports concurrent maximum tracking channel of L1/L5: 90/60.

4.3 Time To First Fix (TTFF)

Table 4-1. TTFF specification

GNSS Constellation	Start Type ⁽¹⁾		TTFF	
GPS + GLONASS + Galileo + BeiDou + NavIC	Cold start		Mean	< 26 s
			Max	< 40 s
	Warm start		Mean	< 26 s
			Max	< 38 s
	Hot start		Mean	< 1 s
			Max	< 3 s
	Aided with EPO ⁽²⁾	Cold start	Mean	< 15 s
		Warm start	Mean	< 5 s
	Aided with A-GNSS ⁽³⁾	Cold start	Mean	< 5 s
		Warm start	Mean	< 3 s
		Hot start	Mean	< 1 s
BeiDou-Only ⁽⁴⁾	Cold start		Mean	< 30 s
			Max	< 40 s
	Warm start		Mean	< 26 s
			Max	< 38 s
	Hot start		Mean	< 1 s
			Max	< 3 s

(1) Satellite signals at -130dBm, HDOP ≤ 1.5 , PDOP ≤ 2.5 , tested at 25°C.

(2) With correct system time or NTP.

(3) Supported constellations depend on service provider.

(4) BDS B1I, B1C, B2a ≥ 6 SVs.

4.4 1PPS Accuracy

MT8668 supports 1PPS with accuracy < 80 ns (95% at 25°C) or < 250ns (95% at -40°C ~ 85°C).

Note:

- Tested with PPS calibration and full set of DCB calibration
- Satellite signals ≥ -140 dBm.

4.5 Positioning Accuracy

Table 4-2. Positioning accuracy

GNSS Constellation	Positioning Accuracy ⁽¹⁾		
GPS + GLONASS + Galileo + BeiDou + NavIC ⁽²⁾	SPP	Horizontal	< 2 m (CEP95)
		Vertical	< 4 m (CEP95)
BeiDou-Only ⁽³⁾	SPP	Horizontal	< 2 m (CEP95)
		Vertical	< 4 m (CEP95)

- (1) Static test scenario with GNSS simulator: Static test for 1 hour under opensky environment (no obstruction above 5° elevation angle), with HDOP ≤ 1.5 , PDOP ≤ 2.5 , and all satellite signals at -130dBm.
- (2) Constellations with at least GPS L1C/A, BDS B1I, B2a, GAL E1, and GLO L1OF. Number of visible satellites with at least 6SV for each constellation.
- (3) BDS B1I, B2a ≥ 6 SVs.

4.6 Velocity Accuracy

Table 4-3. Velocity accuracy

GNSS Constellation	Velocity Accuracy	
GPS + GLONASS + Galileo + BeiDou + NavIC ⁽³⁾	SPP ⁽¹⁾	< 0.5 m/s (CEP95)
	SPP under constant speed ⁽²⁾	< 0.27 m/s (CEP95)
BeiDou-Only ⁽⁴⁾	SPP ⁽¹⁾	< 0.5 m/s (CEP95)
	SPP under constant speed ⁽²⁾	< 0.27 m/s (CEP95)

- (1) Dynamic test scenario with GNSS simulator:
Dynamic test for 1 hour under opensky environment, with HDOP ≤ 1.5 , PDOP ≤ 2.5 , and all satellite signals at -130dBm.
Trajectory settings as follows:
1. Start moving north at 30 km/h;
 2. Accelerate from 30 km/h to 150 km/h within 85 meters;
 3. Maintain 150 km/h for 730 meters, then decelerate to 30 km/h within 85 meters;
 4. Make a 90-degree clockwise turn with a 20-meter radius at 30 km/h;
 5. Accelerate to 150 km/h within 85 meters;
 6. Maintain 150 km/h for 1230 meters, then decelerate to 30 km/h within 85 meters;
 7. Make another 90-degree clockwise turn with a 20-meter radius at 30 km/h;
- Repeat steps 2 to 7 to complete a rectangular (rounded corners) path and continue the same trajectory until the end of the scenario.
- (2) Constant speed test scenario with GNSS simulator:
Under opensky environment (no obstruction above 5° elevation angle), with HDOP < 1.2 , and all satellite signals at -130dBm.
Vehicle speed configuration:
1. 0 ~ 1st minute: Speed = 0 km/h.
 2. 1st ~ 3rd minute: Accelerates from 0 km/h to 40 km/h, 80 km/h, and 120 km/h respectively with fixed acceleration.
 3. 3rd ~ 60th minute: Maintain horizontal speed at 40 km/h, 80 km/h and 120 km/h respectively.
- The statistical result is calculated after the receiver speed reaches 40km/h, 80km/h, 120km/h with at least 10 minutes of constant speed data.
- (3) Constellations with at least GPS L1C/A, BDS B1I, B1C, B2a, GAL E1, and GLO L1OF. Number of visible satellites with at least 6SV for each constellation.
- (4) BDS B1I, B2a ≥ 6 SVs.

4.7 Sensitivity

Table 4-4. Sensitivity

GNSS Constellation	Sensitivity		
GPS + GLONASS + Galileo + BeiDou + NavIC	Acquisition	Cold start	-148 dBm
		Warm start	-151 dBm
		Hot start	-163 dBm
	Tracking		-165 dBm
BeiDou-Only	Acquisition	Cold start	-142dBm
		Warm start	-146dBm
		Hot start	-161dBm
	Tracking		-161dBm

4.8 Temperature

Table 4-5. Temperature

Status	Temperature
Working temperature	-40°C to 85°C (T _a)

4.9 Operation Limits

Table 4-6. Operation limits

Parameter	Operation Limit
Dynamic	< 4 g
Altitude	< 18270 m
Velocity	< 515 m/s

4.10 UTC Format

MT8668 supports UTC format as *hhmmss.mmm*.

4.11 Update Rate

Table 4-7. Update rate

GNSS Constellation	Update Rate	
GPS + GLONASS + Galileo + BeiDou + NavIC	PVT	1/2/5/10 Hz
	RAW	1/2/5/10 Hz
	ADR ⁽¹⁾	1/2/5/10 Hz

(1) The update rate of ADR must be equal to the update rate of PVT.

4.12 NMEA GGA Sentence Latency

MT8668 supports NMEA GGA sentence latency of $95\% < 80\text{ ms}$.

Note:

- This feature supports GNSS only (without ADR).

4.13 ADR EDT

MT8668 supports ADR v4.1 with EDT $< 1\%$.

Note:

- CEP68 with 1000 to 3000 meters of distance travelled.

5 Message Output

5.1 Message Output

Table 5-1. Message output

Category	Message
NMEA 0183 ⁽¹⁾	GGA
	GSA
	GSV
	RMC
	VTG
RTCM	v3.2 ⁽²⁾
Raw measurement data ⁽³⁾	Pseudorange
	Carrier ⁽⁴⁾
	C/N0
	Doppler ⁽⁵⁾
Ephemeris	Ephemeris
Leap second	(By request)

- (1) MT8668 support NMEA 0183 v4.10.
- (2) Partial support, please refer to the RTMC document^[2] for details.
- (3) Android raw measurement format.
- (4) Unit: meter.
- (5) Unit: m/s.

6 Contact

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